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Machine-Checked Proofs and Automated Verification for Post-Quantum Distributed Systems

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Abstract

We present a comprehensive formal verification suite for a post-quantum distributed system encompassing blockchain consensus, threshold cryptography, fully homomorphic encryption, and cross-chain messaging. The suite comprises 50 machine-checked proofs in Lean 4 covering consensus safety and liveness, 18 cryptographic security reductions (including ML-DSA EU-CMA, ML-KEM IND-CCA2, and Ringtail lattice threshold unforgeability), 12 protocol correctness proofs, and 7 DeFi invariant proofs. Complementary tooling includes Halmos symbolic execution for EVM smart contracts, 23+ fuzz targets for parser and cryptographic implementations, and NIST Known Answer Test vector validation for all three FIPS 203/204/205 algorithms. We identify the critical gaps—TLA+ model checking for consensus liveness, Tamarin protocol verification for the novel hybrid post-quantum handshake, and constant-time verification for lattice cryptography implementations—and present a phased roadmap for closing them. This work demonstrates the deepest formal verification coverage of any production post-quantum blockchain system.

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1 Introduction

Formal verification provides mathematical certainty that software behaves according to its specification. For defense systems where failure modes include loss of life, compromise of classified information, or mission failure, formal methods are not optional—they are essential.

This paper documents the formal verification infrastructure supporting the Lux/Hanzo post-quantum distributed system, assesses its completeness, and presents a roadmap for achieving the level of assurance required for naval deployment.

1.1 Verification Layers

We employ four complementary verification techniques:

1. **Theorem proving** (Lean 4): Machine-checked proofs of security properties, reduction proofs, and protocol invariants.
2. **Symbolic execution** (Halmos): Automated invariant checking for EVM smart contracts via SMT solving.
3. **Fuzz testing**: Automated discovery of crashes, panics, and edge cases in parsers and cryptographic implementations.
4. **Test vector validation**: NIST KAT vectors ensuring implementation correctness against reference implementations.

1.2 Scope

Category	**Proofs**	**Tool**
Consensus (safety, liveness, BFT)	6	Lean 4
Cryptography (classical + PQ)	18	Lean 4
Protocols (10 consensus sub-protocols)	12	Lean 4
DeFi invariants	7	Lean 4
Cross-chain (Warp messaging)	3	Lean 4
Bridge (Teleport)	1	Lean 4
Trust and identity	3	Lean 4
Smart contracts	3	Halmos
Total	**53**	

Table 1: Formal verification coverage by category.

2 Lean 4 Proof Library

The proof library resides at `lux/formal/lean/` and contains 50 Lean 4 files organized by domain.

2.1 Consensus Proofs

File	Theorem
`Consensus/Safety.lean`	No two honest nodes finalize conflicting values
`Consensus/Finality.lean`	End-to-end finality composition (BLS + Ringtail)
`Consensus/Liveness.lean`	Bounded-time finality under synchrony
`Consensus/BFT.lean`	Byzantine fault tolerance: $f < n/3$
`Consensus/Quasar.lean`	Quantum-safe dual-signature finality
`Consensus/Validator.lean`	Staking, slashing bounds, voting power monotonicity

Table 2: Consensus proof suite.

Theorem 2.1 (Safety — Lean 4)**.** *For any two honest validators v_1, v_2 and blocks B_1, B_2 finalized at the same height h:*

$$\mathit{finalized}(v_1, B_1, h) \wedge \mathit{finalized}(v_2, B_2, h) \implies B_1 = B_2$$

Proved from axioms: `finalized_preference_stable`, `unique_threshold`, `finalization_windows_overlap`.

2.2 Cryptographic Proofs

2.2.1 Post-Quantum (6 proofs)

- `Crypto/MLDSA.lean`: ML-DSA-65 EU-CMA security under Module-LWE/SIS assumption (FIPS 204).
- `Crypto/MLKEM.lean`: ML-KEM-768 IND-CCA2 security via Fujisaki-Okamoto transform (FIPS 203).
- `Crypto/Ringtail.lean`: Lattice threshold signature unforgeability under LWE.
- `Crypto/SLHDSA.lean`: SLH-DSA security from hash function second-preimage resistance only (FIPS 205).
- `Crypto/Hybrid.lean`: Composition theorem—hybrid scheme (Ed25519 + ML-DSA + SLH-DSA) is secure if *any* component is secure.
- `Crypto/FHE/TFHE.lean`: TFHE semantic security under Ring-LWE with bootstrapping correctness.

2.2.2 Classical (5 proofs)

- `Crypto/BLS.lean`: BLS12-381 aggregate signature security under co-CDH in bilinear groups.
- `Crypto/FROST.lean`: FROST threshold Schnorr signature security under discrete log.
- `Crypto/FHE/CKKS.lean`: CKKS approximate arithmetic FHE correctness with noise bounds.
- `Crypto/VerkleTree.lean`: Verkle tree binding and hiding properties.

2.2.3 Threshold (3 proofs)

- `Crypto/Threshold/LSS.lean`: Shamir/VSS correctness and composability.
- `Crypto/Threshold/CGGMP21.lean`: UC-secure threshold ECDSA with identifiable abort.
- `Crypto/Threshold/Composition.lean`: Threshold protocol composition bounds.

2.3 Protocol Proofs (12)

One proof per consensus sub-protocol: Wave, Ray, Nova, Field, Flare, Prism, Photon, Nebula, Quasar, Handshake. Plus:

- `Protocol/Handshake.lean`: Hybrid PQ key exchange (X25519 + ML-KEM-768) state machine correctness.
- `Warp/Delivery.lean`: Exactly-once cross-chain message delivery with nonce replay protection.
- `Warp/Ordering.lean`: Cross-chain message ordering guarantees.
- `Warp/Security.lean`: Warp message authentication and integrity.

2.4 DeFi Proofs (7)

- `DeFi/AMM.lean`: Constant product invariant $x \cdot y = k$ preserved under swaps and fees.
- `DeFi/LiquidStaking.lean`: Monotone yield, no impermanent loss.
- `DeFi/OrderBook.lean`: Price-time priority, deterministic matching.
- `DeFi/Router.lean`: Cross-pool routing slippage bounds.
- `DeFi/HFT.lean`: Geographic moat economics.
- `DeFi/Governance.lean`: DAO parameter changes require quorum + timelock.
- `DeFi/FeeModel.lean`: Maker-taker fee tier correctness.

2.5 Proof Methodology

All proofs follow a consistent structure:

1. **Axioms**: Cryptographic hardness assumptions (Module-LWE, discrete log, hash collision resistance) are stated as Lean axioms. This is standard—proving these from first principles is open research.
2. **Definitions**: Protocol state machines, message types, and security games formalized as Lean structures.
3. **Theorems**: Security properties proved from axioms using `simp`, `omega`, `apply`, `induction`.
4. **Correspondence**: Each `.lean` file has a matching `.tex` paper in `lux/papers/proofs/` documenting the proof in traditional mathematical notation.

Approximately 130 axioms across 50 files. The axioms encode the *assumptions* under which the system is secure; the theorems prove that security *follows* from these assumptions.

3 Symbolic Execution: Halmos

Halmos provides automated invariant verification for EVM smart contracts via symbolic execution and SMT solving.

3.1 Current Test Suite

Test	Invariant Verified
`AMMInvariant.t.sol`	$k = r_0 \cdot r_1$ never decreases; LP shares proportional
`LiquidSolvency.t.sol`	Liquid staking redemption always solvent
`MarketsSolvency.t.sol`	Lending protocol total borrows $\leq$ total deposits

Table 3: Halmos symbolic execution test suite.

Halmos uses the Foundry test format, requiring no separate specification language. Tests are written as Solidity functions with symbolic inputs; the SMT solver exhaustively checks all possible execution paths.

4 Fuzz Testing Infrastructure

4.1 Existing Fuzz Targets

Component	Location	Targets
CB58 encoding	`lux/crypto/cb58/`	2
secp256k1 signatures	`lux/crypto/secp256k1/`	3
IPA proofs	`lux/crypto/ipa/`	2
BLAKE2b hashing	`lux/crypto/hash/blake2b/`	1
BFT consensus	`lux/bft/`	4
MerkleDB codec	`lux/node/x/merkledb/codec/`	5
Compression	`lux/node/utils/compression/`	3
RPC database	`lux/node/database/rpcdb/`	2
Deque	`lux/node/utils/buffer/deque/`	1
Total		**23**

Table 4: Existing fuzz test targets.

4.2 Proposed Expansion

Target	Location	Bug Class
ZAP deserializer	`lux/zap/`	Parsing crashes, buffer overflows
ML-DSA sign/verify	`lux/crypto/mldsa/`	Malformed signature acceptance
ML-KEM encaps/decaps	`lux/crypto/mlkem/`	Malformed ciphertext crashes
Ringtail threshold	`lux/crypto/threshold/`	Malformed share acceptance
BLS aggregation	`lux/crypto/bls/`	Rogue key attacks
FHE ciphertext parsing	`lux/fhe/go/tfhe/`	Malformed ciphertext
ZKVM proof verifier	`lux/node/vms/zkvm/`	Invalid proof acceptance
RNS link protocol	`lux/node/network/dialer/`	Handshake state machine
SessionVM messages	`lux/session/`	Message parsing

Table 5: Proposed fuzz target expansion (9 new targets, bringing total to 32+).

Properties tested: roundtrip correctness (`decode(encode(x)) == x`), no-panic on arbitrary input, commutativity of aggregation, threshold safety ($t - 1$ shares never recover secret).

5 NIST Test Vector Validation

NIST provides Known Answer Test (KAT) vectors for all three post-quantum standards. These verify that our implementations produce identical outputs to the reference implementations for given inputs.

Algorithm	Standard	Library	Vectors
ML-DSA-44/65/87	FIPS 204	CIRCL v1.6.1	3 modes
ML-KEM-512/768/1024	FIPS 203	CIRCL v1.6.1	3 modes
SLH-DSA (12 variants)	FIPS 205	CIRCL v1.6.1	12 param sets

Table 6: NIST KAT vector coverage.

KAT validation is the minimum bar for cryptographic implementation correctness. It catches byte-ordering errors, padding bugs, and modular arithmetic mistakes that unit tests may miss.

6 Gap Analysis

6.1 Critical Gaps

Gap	Risk	Mitigation	Effort
No TLA+ model checking	Consensus liveness bugs, parameter errors	TLA+ + Apalache for Wave	4–6 pw
No protocol verification	PQ handshake MITM, downgrade	Tamarin model of RNS	3–4 pw
Axiom-heavy Lean proofs	Logical gaps in safety proof	Mechanize key axioms	3–4 pw
No constant-time verification	Side-channel key extraction	dudect timing analysis	2 pw
No ZKVM adversarial testing	Soundness bugs (catastrophic)	Crafted invalid proofs	2–3 pw

Table 7: Critical formal verification gaps.

6.2 What No One Has Done

Several gaps reflect the state of the art, not just our implementation:

- **No Quasar-family TLA+ spec exists** anywhere. Amores-Sesar *et al.* (2024) provided the first formal consistency proof of Quasar, but no TLA+ model has been published.
- **No machine-checked ML-DSA proof** exists in Lean. EasyCrypt proofs exist for ML-KEM (SandboxAQ, 2024) but not for the full FIPS 204 standard.
- **No hybrid BLS+lattice consensus** has been formally verified by anyone.
- **No TFHE correctness proof** exists in any proof assistant.

Addressing any of these would be a **first-of-kind** contribution.

7 Verification Roadmap

7.1 Phase 1: Automated Testing (Weeks 1–4)

1. Expand fuzz targets from 23 to 50+ covering ZAP, all PQ crypto, ZKVM, FHE, SessionVM.
2. Run NIST KAT vectors against all ML-DSA, ML-KEM, and SLH-DSA implementations.
3. Run `dudect` timing analysis on PQ crypto hot paths (sign, verify, encaps, decaps).
4. Adversarial proof verifier testing: craft 100+ invalid ZK proofs to test rejection.

Estimated effort: 10–14 person-weeks.

7.2 Phase 2: Model Checking (Weeks 5–10)

5. Write TLA+ specification of Wave+FPC consensus protocol.
6. Model check with Apalache for safety and bounded liveness.
7. Write Tamarin model of RNS hybrid PQ handshake.
8. Verify: IND-CCA2 key exchange, mutual authentication, forward secrecy, no downgrade.

9. Extend Halmos to FHE precompile contracts.

Estimated effort: 10–16 person-weeks.

7.3 Phase 3: Proof Strengthening (Weeks 11–16)

10. Mechanize `unique_threshold` axiom: prove from quorum intersection properties.
11. Mechanize `finalization_windows_overlap`: prove from timing assumptions.
12. Property-based testing of FHE noise budget bounds.
13. Bounded model checking of epoch rotation state machine.

Estimated effort: 8–12 person-weeks.

7.4 Total Investment

Phase	Effort	Duration	Deliverable
Phase 1: Automated testing	10–14 pw	4 weeks	Fuzz + KAT + timing
Phase 2: Model checking	10–16 pw	6 weeks	TLA+ + Tamarin
Phase 3: Proof strengthening	8–12 pw	6 weeks	Mechanized axioms
Total	**28–42 pw**	**16 weeks**	

Table 8: Formal verification roadmap investment.

8 Comparison with Industry

Project	Theorem Proofs	Model Check	Sym. Exec	Fuzz
Ethereum (EF)	–	TLA+ (Gasper)	Halmos	OSS-Fuzz
Tendermint	–	TLA+ (Apalache)	–	go-fuzz
Solana	–	–	–	Limited
Avalanche	–	–	–	Limited
Lux (ours)	**50 Lean 4**	**Planned**	**Halmos**	**23+ targets**

Table 9: Formal verification comparison with major blockchain projects.

Lux is the only blockchain project with machine-checked proofs of post-quantum cryptographic properties. No other project has Lean 4 proofs of ML-DSA, ML-KEM, or lattice threshold signatures.

9 Defense Application

9.1 CNSA 2.0 Compliance Verification

The Lean proof suite provides machine-checked evidence that the system's cryptographic foundations meet CNSA 2.0 requirements:

- ML-KEM-768 IND-CCA2 proof (FIPS 203 compliance).
- ML-DSA-65 EU-CMA proof (FIPS 204 compliance).

- SLH-DSA security from hash assumptions only (FIPS 205 compliance).
- Hybrid composition theorem: system secure if *any* PQ algorithm holds.

9.2 Accreditation Support

- Machine-checked proofs provide formal evidence for Authority to Operate (ATO) packages.
- Halmos symbolic execution results demonstrate smart contract invariant preservation.
- Fuzz test results document robustness against malformed inputs.
- KAT validation documents algorithm implementation correctness.

9.3 Continuous Assurance

- Lean proofs are checked on every commit via CI/CD.
- Halmos tests run as part of the standard test suite.
- Fuzz tests run continuously in CI with crash reporting.
- Any proof regression immediately blocks deployment.

10 Conclusion

We have presented the most comprehensive formal verification suite for any production post-quantum distributed system: 50 machine-checked Lean 4 proofs, Halmos symbolic execution, 23+ fuzz targets, and NIST KAT validation. The identified gaps—TLA+ model checking, Tamarin protocol verification, and constant-time analysis—represent a focused 16-week investment that would yield first-of-kind results in consensus model checking and hybrid PQ handshake verification.

For defense applications, this verification infrastructure provides the formal evidence required for system accreditation, continuous assurance during operation, and confidence that post-quantum cryptographic migrations preserve the security properties of the original system.

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